

Decision trees

INF-GRAM — Machine Learning for Healthcare

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Session roadmap



Why study decision trees?

Decision trees are useful because they are:

- intuitive to explain;
- capable of modelling non-linear effects;
- able to capture interactions automatically;
- the basic building block of many ensemble methods.

Learning objectives

By the end of the lecture, you should be able to:

- explain how a decision tree makes a prediction;
- understand how a split is chosen;
- interpret impurity reduction;
- identify the main hyperparameters controlling complexity;
- explain why trees are powerful but unstable.

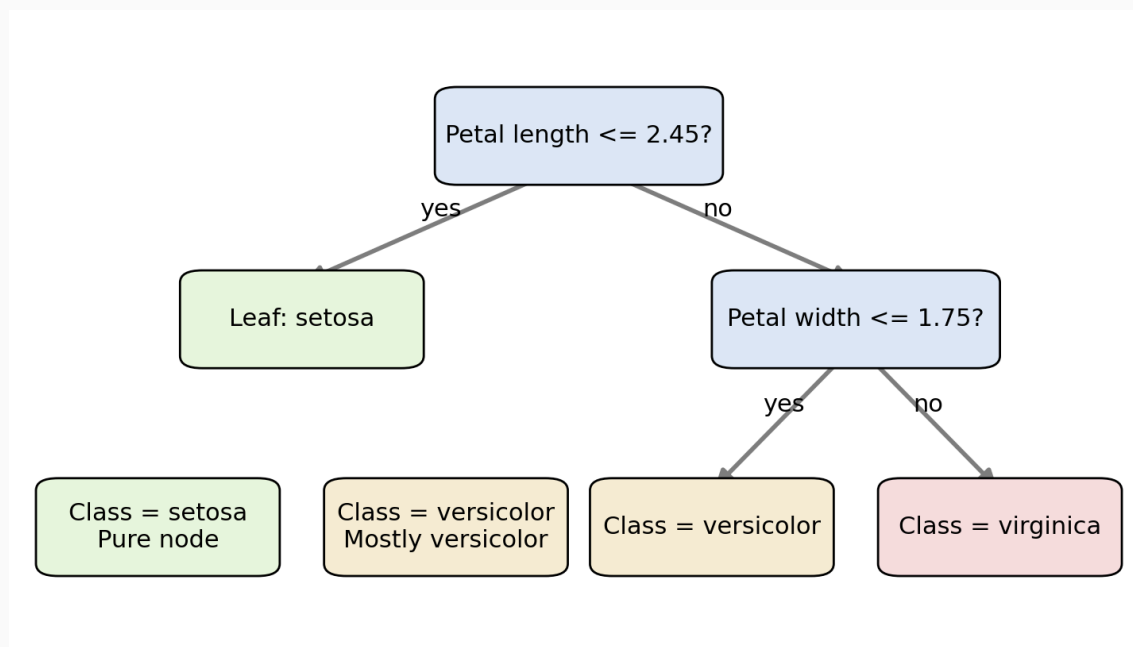
How a tree makes predictions

A tree is a sequence of simple questions

At each internal node, the model asks a yes/no question such as:

- petal length ≤ 2.45 ?
- age < 70 ?
- biomarker > 12.3 ?

By following the path, we reach a **leaf** that contains the final prediction.



Why trees are attractive

A decision tree is easy to read because it mirrors a decision process:

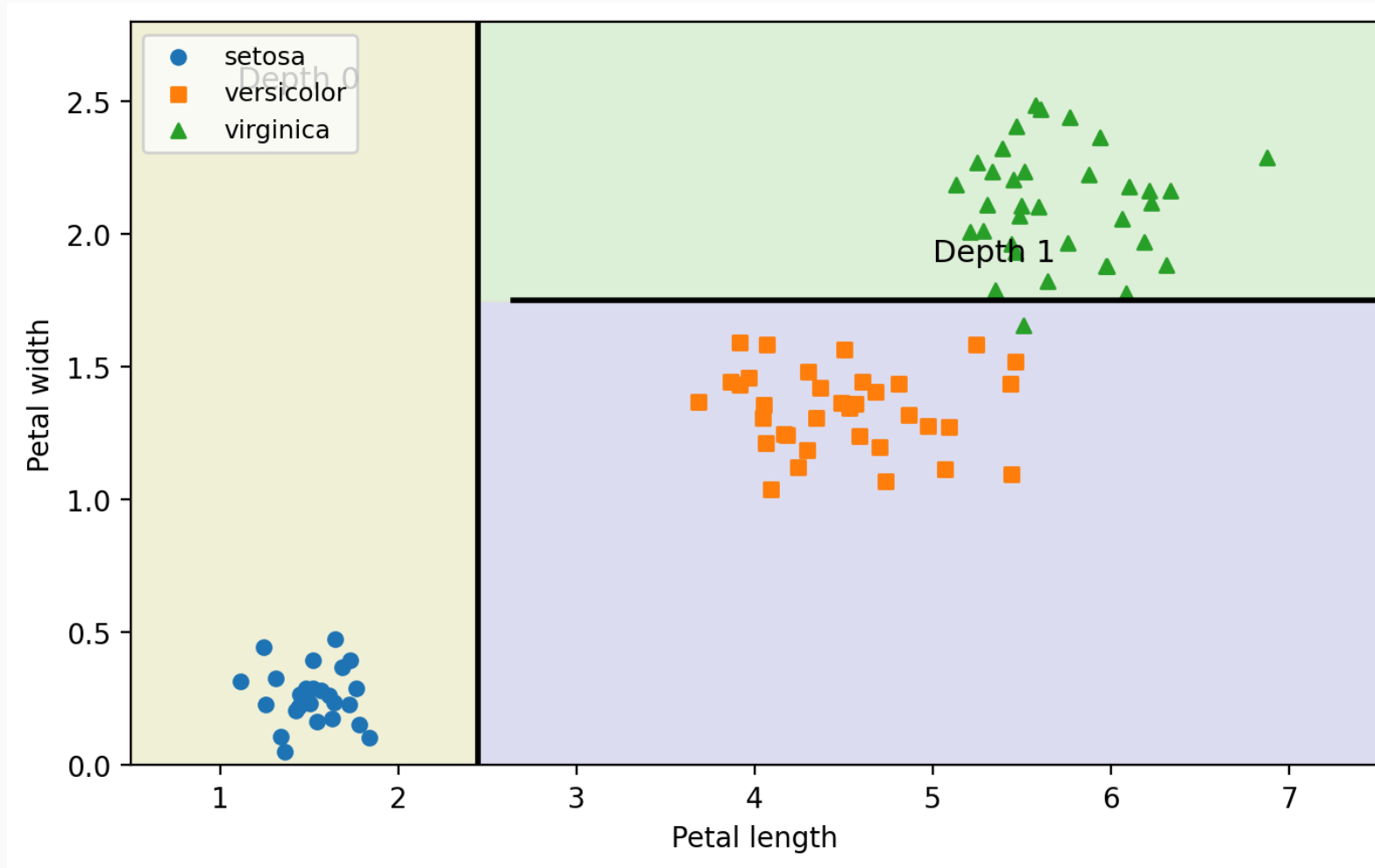
- start from the root;
- apply a sequence of rules;
- end in a leaf with a class or a numerical prediction.

This makes trees a natural introduction to non-linear models.

Splitting the feature space



A tree partitions the space into regions



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Each split creates simpler, more homogeneous regions.

What makes a “good” split?

A good split should send similar observations to the same child node.

In classification, this means reducing **class impurity**. Common criteria are:

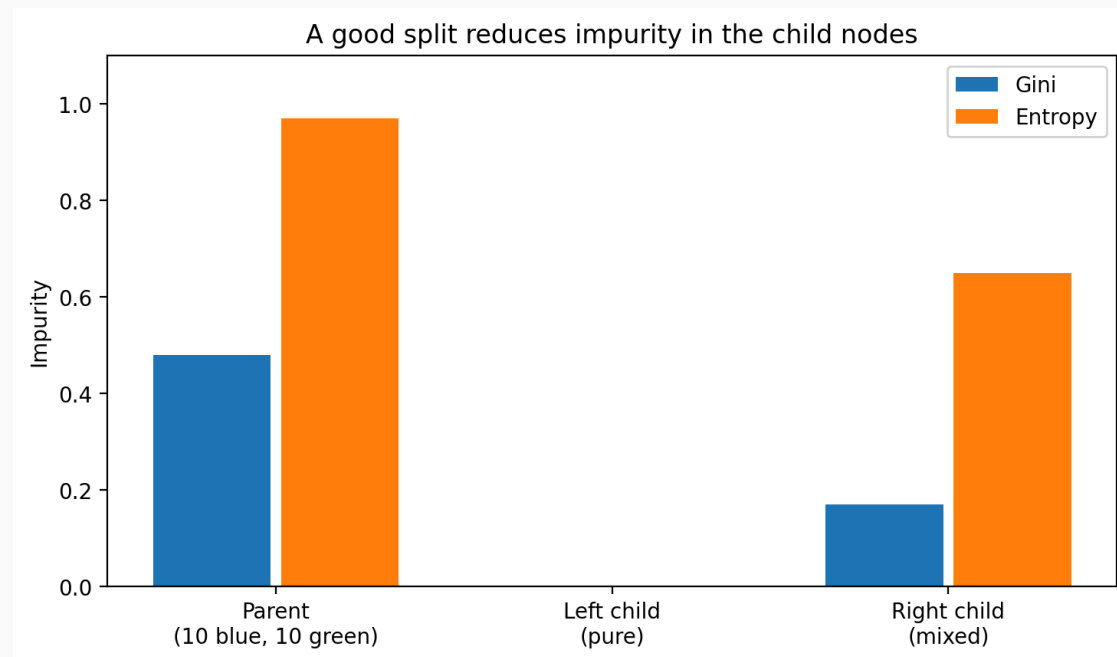
- **Gini impurity**;
- **entropy**.

Visualizing impurity reduction

A parent node is mixed.

After a good split:

- one or both children become purer;
- the weighted impurity decreases;
- the tree becomes more informative.



One simple formula to keep in mind

A split is selected because it improves the node quality:

$$[= - \sum_{k \in \{\text{left}, \text{right}\}} w_{k, _k}]$$

The split with the largest gain is chosen.

Growing the tree... and stopping it



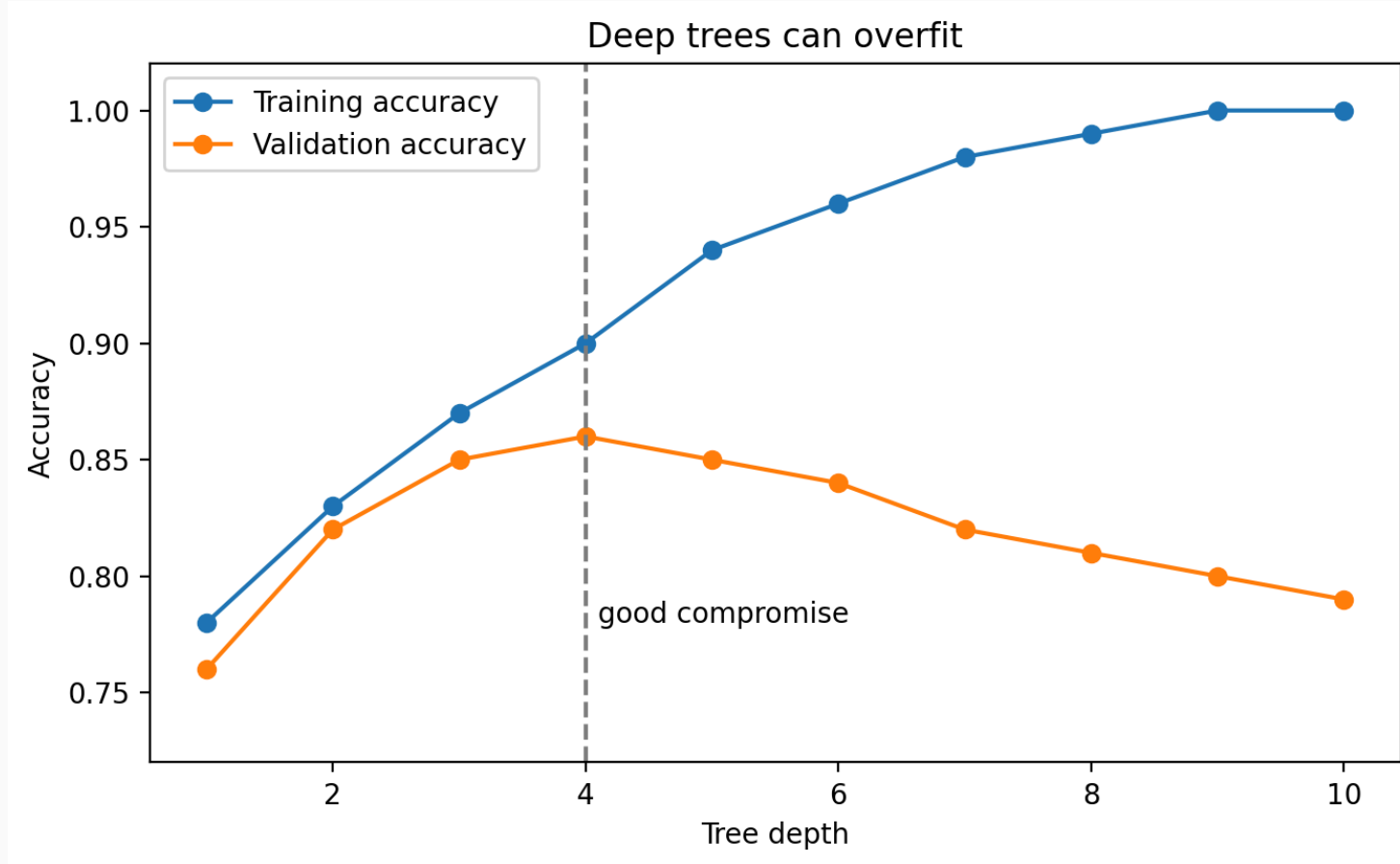
Why not keep splitting forever?

If we continue splitting without restraint, the tree can become:

- very deep;
- extremely specific to the training sample;
- unstable on new data.

This is the typical mechanism of **overfitting**.

Depth and overfitting



The main hyperparameters

To control tree complexity, we tune for example:

- `max_depth`;
- `min_samples_split`;
- `min_samples_leaf`;
- `max_leaf_nodes`.

They all aim to limit unnecessary fragmentation of the data.

Strengths and limitations

Why trees are useful

Strengths:

- easy to communicate;
- work for classification and regression;
- capture interactions automatically;
- require little preprocessing;
- can handle non-linear relationships.

Why a single tree is often not enough

Limitations:

- high variance;
- sensitivity to small data changes;
- tendency to overfit if grown too deeply;
- sometimes lower predictive performance than ensembles.

This is why random forests and boosting methods are so important.

Take-home messages

What to remember from this lecture

- A decision tree is a sequence of binary rules leading to a prediction.
- The algorithm chooses splits that reduce impurity.
- Hyperparameters control complexity and prevent overfitting.
- Trees are intuitive and flexible, but often unstable when used alone.